

Development Automation **& Linux Lab**

Experiment 1

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EXPERIMENT-1

Introduction to Linux Commands

1. Write brief description of the following Shell commands. Also, attach screenshot after running these commands on the terminal.

- man: an interface to the system manuals.

```
MAN(1)                                Manual pager utils                                MAN(1)

NAME
    man - an interface to the system reference manuals

SYNOPSIS
    man [man options] [[section] page ...] ...
    man -k [apropos options] regexp ...
    man -K [man options] [section] term ...
    man -f [whatis options] page ...
    man -l [man options] file ...
    man -w|-W [man options] page ...

DESCRIPTION
    man is the system's manual pager. Each page argument given to man is normally the name of a program,
    utility or function. The manual page associated with each of these arguments is then found and dis-
    played. A section, if provided, will direct man to look only in that section of the manual. The de-
    fault action is to search in all of the available sections following a pre-defined order (see DE-
    FAULTS), and to show only the first page found, even if page exists in several sections.

    The table below shows the section numbers of the manual followed by the types of pages they contain.

    1 Executable programs or shell commands
    2 System calls (functions provided by the kernel)
    3 Library calls (functions within program libraries)
    4 Special files (usually found in /dev)
    5 File formats and conventions, e.g. /etc/passwd
    6 Games
    7 Miscellaneous (including macro packages and conventions), e.g. man(7), groff(7), man-pages(7)

Manual page man(1) line 1 (press h for help or q to quit)
```

- whoami: prints the user name associated with the current effective user ID.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ whoami
ayroid
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- pwd: prints name of the current working directory.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ pwd
/mnt/c/college/upes/sem 4/linux/lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- mkdir: used to create new directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
Experiments  Lab File  Notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ mkdir newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
Experiments  Lab File  Notes  a.txt  expl.txt  newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- cd: used to change directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ pwd
/mnt/c/college/upes/sem 4/linux/lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ cd notes
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/notes$ pwd
/mnt/c/college/upes/sem 4/linux/lab/notes
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/notes$
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/notes$ pwd
/mnt/c/college/upes/sem 4/linux/lab/notes
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab/notes$ cd ..
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ pwd
/mnt/c/college/upes/sem 4/linux/lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls: used to list the contents of the directory.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
Experiments  Lab File  Notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls -l: used to list the contents of the directory in a long listing format.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -l
total 0
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:35 Experiments
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:43 Lab File
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:25 Notes
-rwxrwxrwx 1 ayroid ayroid 81 Jan 17 23:54 a.txt
-rwxrwxrwx 1 ayroid ayroid 151 Jan 20 10:27 expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls -a: shows all contents of the directory + entries starting with .

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux$ ls -a
.  ..  lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux$
```

- ls -c: sorts and shows the content of the directory according to name/time when combined with l/lt respectively.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -c
Lab File  Experiments  Notes  expl.txt  a.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -cl
total 0
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:35 Experiments
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 19:29 Lab File
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:25 Notes
-rwxrwxrwx 1 ayroid ayroid 81 Jan 17 23:54 a.txt
-rwxrwxrwx 1 ayroid ayroid 151 Jan 20 10:27 expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -clt
total 0
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 19:29 Lab File
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:35 Experiments
drwxrwxrwx 1 ayroid ayroid 4096 Jan 26 18:25 Notes
-rwxrwxrwx 1 ayroid ayroid 151 Jan 20 10:27 expl.txt
-rwxrwxrwx 1 ayroid ayroid 81 Jan 17 23:54 a.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls -m: shows the content of the directory separated by commas

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -m
Experiments, Lab File, Notes, a.txt, expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls -p: shows the content of the directory with / indicating directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -p
Experiments/ Lab File/ Notes/ a.txt expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ls -r: shows the content of the directory by sorting them in reverse

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -r
expl.txt a.txt Notes Lab File Experiments
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

ls -f: shows the content of the directory without sorting or colouring them

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls -f
.  ..  a.txt  expl.txt  Experiments  Lab File  Notes
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- clear: clears the terminal screen

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt  newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ rmdir newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ clear
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- rmdir: removes empty directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt  newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ rmdir newdir
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- touch: creates new files without opening the text editor

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ touch b.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  b.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- cat: concatenates files and prints content on the screen

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ cat expl.txt
man
whoami
ls (a, l, m, c, p, r)
ls -l
ls -a
```

- less: shows file's contents one screen at a time. It doesn't load the entire file but accesses it page by page.

```
1
2
3
4
5
6
7
8
9
10
```

- more: used to view the first section of the file.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ more c.txt
1
2
3
4
5
6
7
8
9
10
```

- rm: removes files and directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  b.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ rm b.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments  lab file  notes  a.txt  expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- sort: sorts lines of the text files

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ sort a.txt
1
10
11
12
13
14
15
16
17
18
19
2
```

- cp: copies files and directories

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls ..
lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments lab file notes a.txt expl.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ cp a.txt /mnt/c/college/upes/'sem 4'/linux
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls ..
lab a.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- mv: moves and renames files

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments lab file notes a.txt expl.txt scripts
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls ..
lab
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ mv a.txt /mnt/c/college/upes/'sem 4'/linux
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls ..
lab a.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments lab file notes a.txt expl.txt scripts
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ mv a.txt c.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ ls
experiments lab file notes c.txt expl.txt scripts
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- wc: prints newline, word, and byte counts for each file

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ wc c.txt
30 30 81 c.txt
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- head: prints first n lines of the file. The default value of n is 10.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ head c.txt
1
2
3
4
5
6
7
8
9
10
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- tail: prints the last n lines of the file. The default value of n is 10.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ tail c.txt
21
22
23
24
25
26
27
28
29
30
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- history: GNU history library. Keeps record of the previously used commands.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ history
1  ls
2  cd ..
3  ls
4  cd home
5  ls
6  cd ayroid
7  ls
8  cd ..
9  ls
10 cd ..
```

2. Write in details about the grep command.

grep searches for PATTERNS in each FILE. PATTERNS is one or more patterns separated by newline characters, and grep prints each line that matches a pattern. Typically, PATTERNS should be quoted when grep is used in a shell command.

- i) Find all the patterns that start with “ep” in a given file.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep ep new.txt
ep
episode
hepi
episodes
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- ii) Print those lines of a given text file which do not contain the word “hi”.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep -v hi new.txt
ep
episode
hahah
hepi
episode
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- iii) Explain the usage of options -i, -A, -B, -C in grep command with examples.

- grep -i: ignores case while looking for a match.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep -i "hi" new.txt
hi
Hisikes
phiness
KrHies
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$
```

- grep -A n: prints the matched results and n number of lines after that.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep -A 1 "hi" new.txt
hi
ep
--
phiness
KrHies
```

- grep -B n: prints the matched results and n number of lines before that.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep -B 1 "hi" new.txt
hi
--
Hisikes
phiness
```

- grep -C n: prints the matched results and n number of lines before & after that.

```
ayroid@Ayroid:/mnt/c/college/upes/sem 4/linux/lab$ grep -C 1 "hi" new.txt
hi
ep
--
Hisikes
phiness
KrHies
```